

Two researchers independently read through these responses and marked responses that were likely AI-generated. Any disagreement was reviewed by a third researcher. In the end, we marked 7 participants as likely having used AI to generate open-text responses throughout the study and removed their data from analyses. We thus conducted a second round of data collection and applied the same screening criteria to attain a larger sample size with sufficient power for data analysis. We targeted having at least 5 readers rating each passage; the final sample size after Round 2 of data collection is $N = 30$. For all analyses, we reported results using data from both rounds of data collection.

4 RESULTS

4.1 Overview of Data Analysis

We conducted analyses with three streams of data: a) the interview data and b) writing logs (i.e., behavioral data) from writers in Part 1, and c) survey data from readers in Part 2. We transcribed the interview data and analyzed the transcripts through thematic coding and affinity diagrams. The first author led the qualitative data analysis, and met the research team weekly to discuss findings in order to reduce subjectivity during the data analysis phase (see Appendix C for the full list of emerging themes). When reporting findings from our qualitative data, we follow [1] and use the following phrases to indicate the portion of our writer participants who shared each insight: a few (1-5), some (6-10), most (11-15), nearly all (16-19).

For any within-subject comparison (e.g. writers' behavioral logs using the two versions of the AI tool), we used linear mixed effect models, controlling for participants' subject IDs and order effect.³ For scales that capture individuals' opinions toward AI-assisted writing (e.g., readers' survey responses), we used Wilcoxon one-sample tests to examine whether participants' ratings differ significantly from the midpoint (3) of the five-point scales.⁴ Details for our analytic approaches are listed in Table 5. We organize our results according to our research questions with data from different streams, when applicable. We also present two additional points that emerged from our interviews with writers:

- In Section 4.2, we first address writer-centered definitions of authenticity in writing and the impact of co-writing with AI on authenticity (RQ1).
- In Section 4.3, we elaborate on how writers would like to preserve their authentic voices through practices of writing with AI (RQ2).
- In Section 4.4, we compare whether and how personalization of AI writing assistance affects writers' and readers' experiences, respectively (RQ3).
- As discussions around writer-reader relationships as well as alternative forms of writing support emerged frequently in our interviews with writers, we further elaborate on them in Section 4.5 and Section 4.6, respectively.

4.2 Writer-Centered Conceptions of Authenticity in Writing

Our findings first reveal that writers conceptualize authenticity through *the source of content, internal experiences and identities* that ground their work, and the actual *writing outcomes*. Although participants' reflections did resonate with some of the definitions of authenticity as established in the existing literature (i.e., category, source, and value), they placed further emphasis on viewing authenticity through their internal experiences in addition to through their explicit expressions in writing. Moreover, regardless of how participants defined and understood authenticity, many of them indicated that authentic writing *is* the essence of good writing, and saw the likely impact of AI on authenticity in

³To run the linear mixed effect models, we used the *|lmer|* R package.

⁴We used Wilcoxon one-sample tests instead of normality tests for distribution as each of these variables was measured by a single 5-point scale. Thus, we treated the data as ordinal instead of continuous numeric values.

Dependent variable	Test goal	Model tested
Part 1: Writers' Behavioral Data		
Frequency of requesting AI assistance; Acceptance rate of AI suggestions	To compare behavioral patterns of each writer in the personalized vs. non-personalized condition	$\text{lmer}(\text{DV} \sim \text{condition} + \text{session order} + (1 \text{writerID}), \text{data})$
Part 2: Readers' Self-report Data		
Familiarity with generative AI; Interest in reading AI-assisted writing; Authenticity of AI-assisted writing	To understand readers' existing perception toward AI-assisted writing by examining the distribution of each variable	$\text{wilcox.test}(\text{DV}, \mu = 3, \text{conf.int}=\text{TRUE})$
Likeability, enjoyment, and creativity of writing; Likelihood of human writing	To compare each reader's perception after reading the three writing passages (writers' solo work, work co-written with personalized AI, and work co-written with non-personalized AI)	$\text{lmer}(\text{like} \sim \text{condition} + \text{reading order} + (1 \text{authorID}) + (1 \text{readerID}), \text{data})$
Degree of preserving writers' authentic voices; Credits and authorship of work	To examine how each reader compare the work co-written with personalized AI vs. non-personalized AI to a writer's solo work respectively	$\text{lmer}(\text{like} \sim \text{condition} + \text{reading order} + (1 \text{authorID}) + (1 \text{readerID}), \text{data})$
Perception of the writing; Perception of the human writer; Appreciation and evaluation of writing	To understand readers' perception toward AI-assisted writing after reading AI-co-written work by examining the distribution of each variable	$\text{wilcox.test}(\text{DV}, \mu = 3, \text{conf.int}=\text{TRUE})$

Table 5. Statistical models used for quantitative data analyses

writing. We provide summaries and quotes of each participant's conceptions of authenticity in Table 6 and further elaborate on the three key themes of writer-centered definitions of authenticity from Sections §4.2.1 to §4.2.3.

4.2.1 Defining authenticity through the source of content (Source authenticity). Several writers described authenticity as a "who" question, focusing on who wrote the text or was the source of the content. This concept mirrors *Source*, a long-standing theoretical component of authenticity in the existing literature, and is directly related to both writers' and readers' considerations for authorship. Participants whose conceptualizations aligned with *Source* authenticity emphasized the *entity* who took actions and contributed to a piece of work. For example, such actions might include producing a piece of text or trying to understand the audience's interest.

Writers who held this view also saw AI writing assistance as a direct threat to authenticity. With AI participating in the writing process, writers are no longer the sole source of content generation, raising questions about the authenticity of their writing. A few participants highlighted the importance of having writers produce their work themselves, as they learn, revise, and refine their work through the actions and processes of writing. Furthermore, they also believed that audiences learn more about writers through the way they produce content.

4.2.2 Defining authenticity through constructing and expressing their authentic self (Authentic self). Writers mentioned several internal states during their writing processes as key constructs of authenticity, many of which have been less covered by prior literature. These include (1) whether writers can freely express their emotions to form emotional connections with their readers, (2) whether the process of writing allows them to feel passionate about their work, (3) whether they have the autonomy to select topics and content that are personally important to them, (4) and most importantly, whether a writer can justify having their name and identity behind their work. In other words, a writer claims the authenticity of their writing if they can soundly argue why the piece of work can only be done by them as *the writer*.

This view of authenticity is shaped by the belief that writers' identities, backgrounds, and lived experiences serve as fundamental materials of writing and enrich their work. These cornerstones—whether writers can genuinely express their own experiences and identities through the writing as well as whether they feel enthusiastic and believe in the

importance of their work—jointly contribute to authenticity. As such, the work would never be in its current form if produced by any other writer. One writer described it metaphorically:

“I’ve often said that writing is a lot like a tree. The trunk of the tree is the idea. Everybody can have the same idea and then start to write something. But eventually, once you get up into the branches, everybody’s going to go off in their own different branch. Authenticity is your own personal branch, the way that you would

Table 6. Writer participants’ definitions of authenticity. “Source”, “Content”, and “Value alignment” are concepts of authenticity identified in prior literature. Construction and expression of authenticity are additional concepts pointed out by our writer participants.

Writer	Summary of participant’s definition of authenticity	Quote	Source authenticity (§4.2.1)	Authentic self (§4.2.2)	Content authenticity (§4.2.3)	Value alignment
P1	Authenticity concerns whether a piece of work is produced by and sounds like its author	<i>“To me, authenticity is ensuring that my own ideas are conveyed through the writing, rather than repeating or summarizing another’s. [...] Did I come up with the idea? Is this unique? Has it been done before? [...] I want to feel like it’s written by [P1’s name].”</i>	✓		✓	
P2	Authenticity concerns whether a piece of work is produced by and sounds like its author	<i>“I only consider if [authenticity] in terms of ‘If you’re starting to sound too much like this other writer’ [...] I guess it’s harder to come up with an original idea just because there’s only so many ideas in the world. But an authentic voice is always possible.”</i>	✓		✓	
P3	Authenticity concerns the human and their living experiences behind the story they wrote.	<i>“Can they get inspiration from AI? Absolutely. But authentic writing is about the story that’s told from the heart of an author. If there is no human behind the story [...] their experiences, their emotions, their insights, and their imagination behind the story, it wouldn’t be as meaningful. And writing would just be pieces of words.”</i>		✓		
P4	Authenticity concerns whether the author’s beliefs align with their work and whether they feel comfortable having their names on and representing themselves through their work.	<i>“Authenticity refers to original work that accurately represents what the author means to convey. What they’re putting out should have their full interest. [...] So I think authenticity has to refer to whether the author could feel strongly standing behind their own work. Whether they feel comfortable having their names behind the work.”</i>		✓		✓
P5	Authenticity concerns who produces the work and puts effort into building connection with their readers.	<i>“Authentic writing is raw, it creates a human connection between two people: the author and reader. The author builds their soul into their work and trusts the reader to understand them. [...] Authenticity is defined by the source of who produces the writing.”</i>	✓	✓		
P6	Authenticity concerns whether the work uniquely represents the author. It is done through the writer’s personal touch that is shaped by their experience and worldview.	<i>“Authenticity is originality. It’s like a fingerprint. It is what makes content unique. As a writer, it’s that unique perspective and personal touch every writer brings to a particular topic. [...] Authenticity represents one’s personal approaches to their work, which is very much based on their life experiences and understanding of the world.”</i>		✓	✓	
P7	Authenticity represents the writer’s unique voice. This is done through expressing the writer’s own experiences and sentiment.	<i>“It’s a certain individual nuance that’s just not the same as anyone else. [...] I make it [authenticity] by adding in my own sensibilities. It’s the expression of myself and my own emotions.”</i>		✓	✓	
P8	Authenticity concerns the writer’s own unique voice and style in writing	<i>“I believe authenticity in writing is dependent on the writer’s ability to present their ideas in their own writing voice, which is completely unique to the individual. To me, as a writer, your voice is the same as your fingerprint. It is what defines you as a writer.”</i>			✓	
P9	Authenticity concerns whether the author can express himself/herself freely through their writing.	<i>“Authenticity in writing means being able to express one’s true self [...] just that I would be able to be myself in my own writing.”</i>		✓		
P10	Authenticity is constructed through the living environments and stories of the writer.	<i>“Authenticity in writing means to maintain the original stories and voice of the writer. [...] A large part of authenticity comes from the experience and environment that the creators are situated in.”</i>		✓		
P11	Authenticity means building connections with the audience. The writer approached this by speaking genuinely to their readers.	<i>“To me, authenticity means incorporating my personal voice. I want them to feel like I am talking to them [readers], not at them. [...] I think just being real and talking to the audience like a person. More just trying to form those relationships and have that more personal connection, put a little personality on when it’s appropriate.”</i>	✓	✓		
P12	Authenticity concerns writing about subjects that represent one’s interest and soul, which makes their work unique from others.	<i>“I have ghostwritten articles about many topics. I was aware of my inauthenticity [in] writing about subjects for which I lacked feeling. After several articles in these lanes, I could not find the vocabulary to make a difference from my fellow writers. [...] I want sounds and words and sentences to convey my passion, my thoughts, my soul.”</i>		✓	✓	